

Exploratory Data Analysis (EDA)

Dataset:

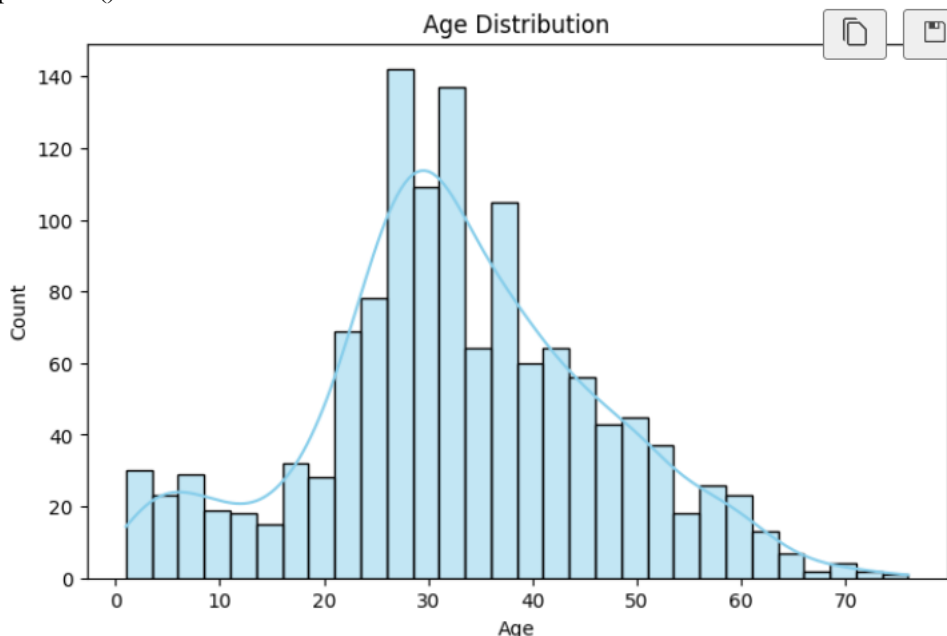
The dataset used is named LusitaniaManifest.csv and appears to include information about passengers aboard the RMS Lusitania. Each row corresponds to an individual record, likely containing attributes such as name, age, gender, nationality, etc.

Data Pre-processing:

EDA:

- 1) This Chart shows how ages are distributed across the population with both the histogram and the KDE curve for a smoother representation of the distribution.

```
plt.figure(figsize=(8, 5))
sns.histplot(data['Age'].dropna(), bins=30, kde=True, color='skyblue')
plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Count")
plt.show()
```



Peak Around Age 28–35:

The tallest bars are around the late 20s to early 30s, suggesting most passengers were young adults.

Slight Left Skew:

The curve is slightly left-skewed, meaning there are more younger passengers than older ones.

Fewer Children and Seniors:

There are relatively fewer children (0–15 years) and elderly people (60+ years) in the dataset.

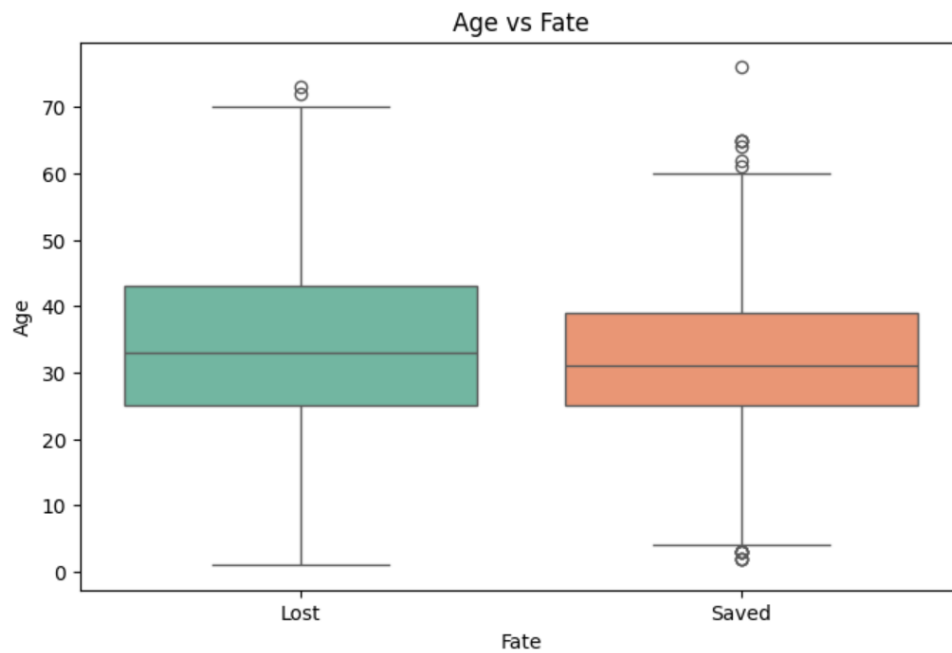
Normal-like Distribution:

Although not perfectly symmetric, the KDE curve somewhat resembles a bell curve, indicating that the dataset may be modelled well by statistical techniques assuming normality.

- 2) Plot used to compare how the distribution of age differs depending on fate.

```
plt.figure(figsize=(8, 5))
sns.boxplot(x='Fate', y='Age', data=data, palette='Set2')
plt.title("Age vs Fate")
plt.xlabel("Fate")
```

```
plt.ylabel("Age")
plt.show()
```



Median Age:

- The black line inside each box is the median age.
- The median age is slightly lower for those who were saved, meaning younger people had a slightly higher chance of survival.

Interquartile Range (IQR):

- The box shows the middle 50% of the data (from 25th to 75th percentile).
- The age range of saved individuals is slightly narrower, indicating more consistency in the ages of survivors.

Outliers:

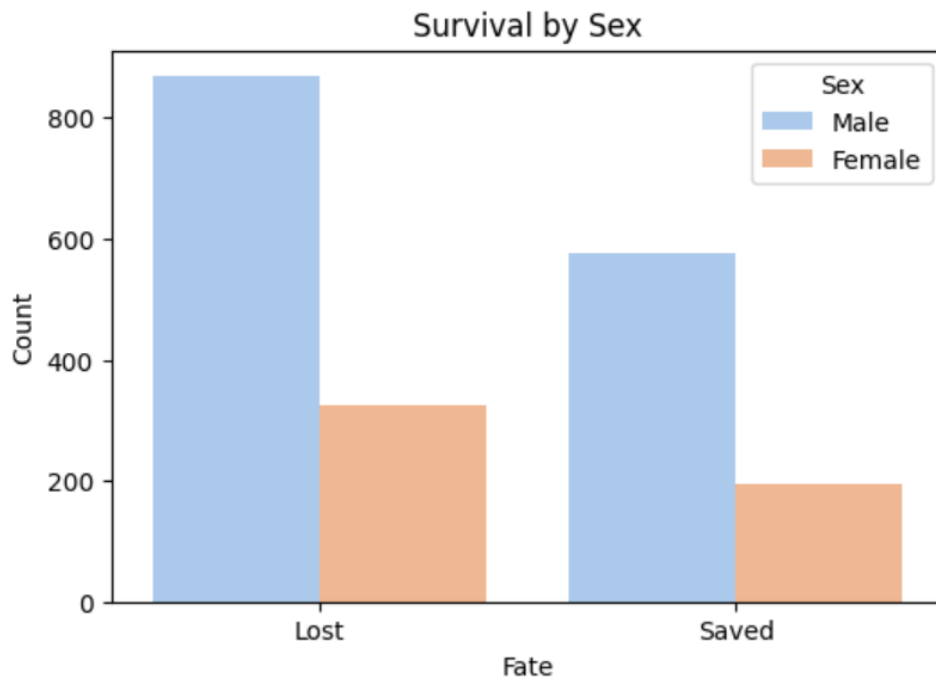
- Small circles above and below the whiskers represent outliers ages that are unusually high or low compared to the rest.
- A few elderly individuals (above 65) were saved,

Age Range:

- Both groups span a wide age range from children (~1–5 years) to the elderly (~70+).
- However, the Lost group has more elderly passengers, and some extreme old ages only appear in this group.

3) This plot quickly highlights gender-based survival trends. It Helps identify if sex had a role in survival outcomes which could be important for further analysis or model feature selection.

```
plt.figure(figsize=(6, 4))
sns.countplot(x='Fate', hue='Sex', data=data, palette='pastel')
plt.title("Survival by Sex")
plt.xlabel("Fate")
plt.ylabel("Count")
plt.show()
```



Overall Survival Pattern:

- More **males** were both **lost** and **saved** compared to females which may reflect that more men were on board overall.

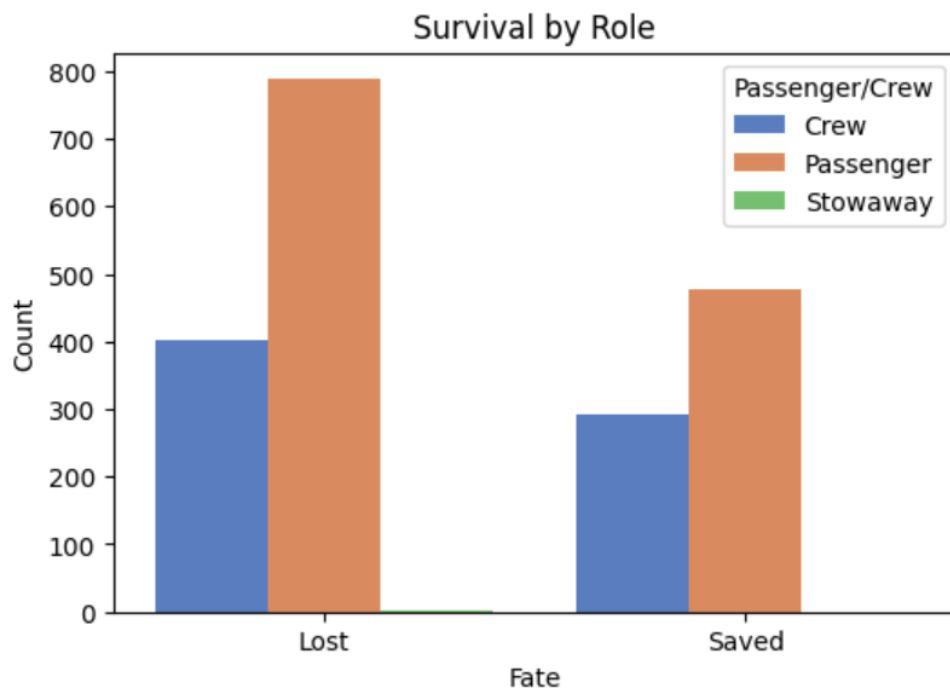
Higher Male Casualties:

- The "**Lost**" bar for **males** is the **tallest**, meaning **most of the casualties were men**.
- Far fewer women were lost in comparison.

Better Survival Rate for Women:

- Although fewer in number, **females have a proportionally higher survival rate**.
 - This could reflect social practices like "**women and children first**" during evacuation (if this was a historical ship disaster like Lusitania or Titanic).
- 4) **This Plot highlights the risk level associated with different roles on the ship, also useful for understanding evacuation priorities or systemic inequalities (e.g., were crew/passengers treated differently during rescue?).**

```
plt.figure(figsize=(6, 4))
sns.countplot(x='Fate', hue='Passenger/Crew', data=data, palette='muted')
plt.title("Survival by Role")
plt.xlabel("Fate")
plt.ylabel("Count")
plt.show()
```



Passengers Were Most Affected:

- The **highest loss** was among **Passengers**—almost **800 passengers were lost**, making up the majority of fatalities.
- This may be due to their location on the ship (e.g., lower decks or crowded conditions).

Crew Had Better Survival Rate:

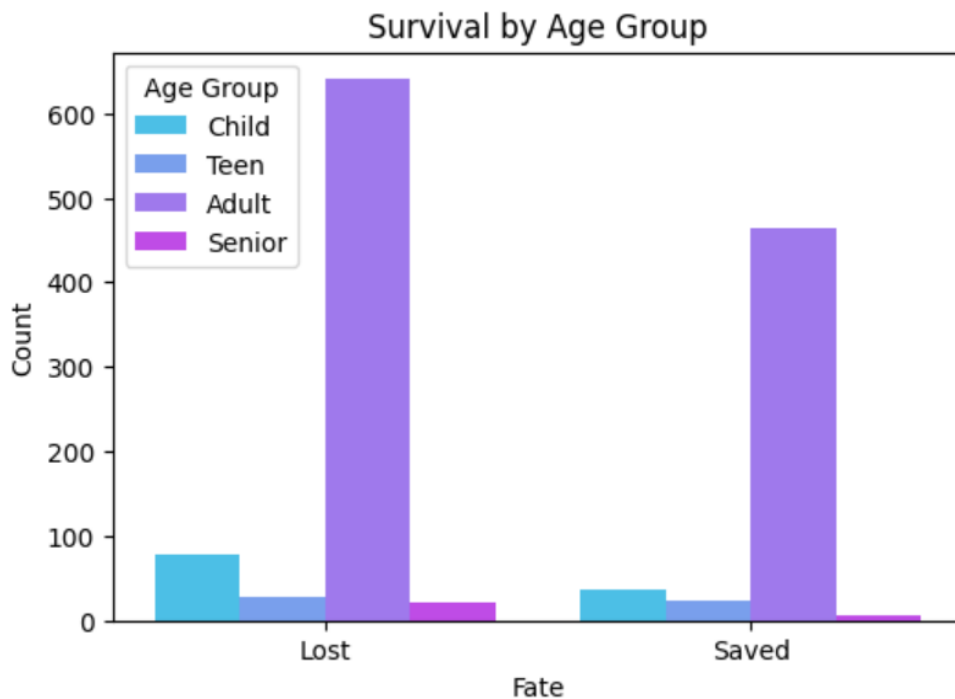
- While many **crew members were lost**, a **significant number survived**—possibly due to training, assigned evacuation duties, or access to exits.
- There's a more balanced distribution between lost and saved among crew compared to passengers.

Stowaways Are Rare and Vulnerable:

- **Very few stowaways** are present in the dataset (barely visible green bar).
- Their count is **extremely low**, and nearly all were **lost**, indicating they were likely the most vulnerable group with **no official access to safety resources**.

5) This plot clearly shows how age impacted survival probability, also helps in understanding rescue behavior, demographic risks, and prioritization during evacuation. This plot offers insight for predictive modeling Age Group can be a strong feature in survival prediction.

```
plt.figure(figsize=(6, 4))
sns.countplot(x='Fate', hue='Age Group', data=data, palette='cool')
plt.title("Survival by Age Group")
plt.xlabel("Fate")
plt.ylabel("Count")
plt.show()
```



Adults Make Up Most of the Data:

- The **Adult** category has the highest counts in both “Lost” and “Saved” outcomes.
- A large number of adults were **lost (~650)**, but a significant number were **saved (~470)** as well.
- This reflects that **adults were the largest demographic onboard**, which skews the overall survival numbers.

Children Had a Slightly Better Chance of Survival:

- While many **children were lost**, the **gap between lost and saved is smaller** compared to other groups.
- This could suggest that **children were given priority** during evacuation efforts (e.g., “women and children first” policy).

Teens Have Low Counts but Follow Similar Trends:

- Teenagers are a **small portion** of the dataset.
- Slightly more teens were lost than saved, but the numbers are relatively close.

Seniors Were Most Vulnerable:

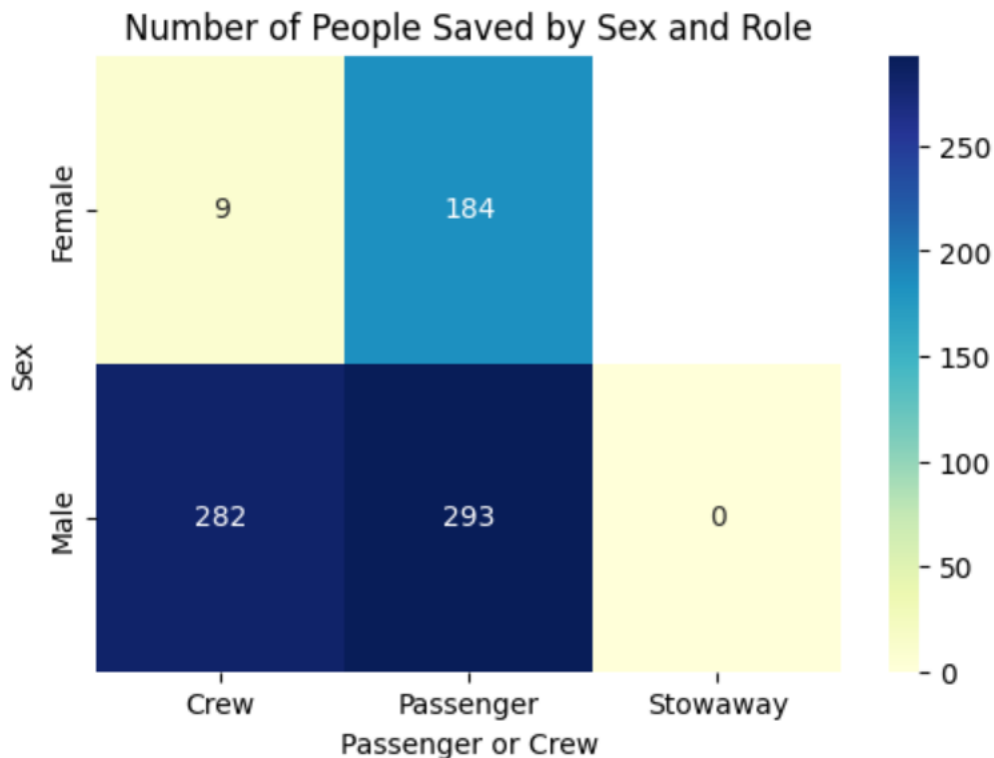
- **Very few seniors survived**, and almost all are in the “Lost” category.
- This reflects the **increased vulnerability** of older individuals in emergencies possibly due to mobility, health conditions, or lack of priority in rescue operations.

6) **This plot highlights disparities in survival by gender and role. Visual cue (colour) quickly shows high vs. low values. Supports analysis of survival patterns and rescue prioritization.**

```
heatmap_data = data.pivot_table(
    index='Sex',
    columns='Passenger/Crew',
    values='Fate',
    aggfunc=lambda x: (x == 'Saved').sum()
)
```

```
plt.figure(figsize=(6, 4))
sns.heatmap(heatmap_data, annot=True, fmt='.0f', cmap='YlGnBu')
plt.title("Number of People Saved by Sex and Role")
plt.ylabel("Sex")
```

```
plt.xlabel("Passenger or Crew")
plt.show()
```



Passengers:

- **Male Passengers Saved:** 293
- **Female Passengers Saved:** 184
- More **male passengers** were saved than females, which is surprising given the common belief of prioritizing women in rescue scenarios.

Crew:

- **Male Crew Saved:** 282
- **Female Crew Saved:** 9
- An overwhelming majority of the saved crew were **males**, suggesting either:
 - Fewer female crew members were onboard.
 - Female crew members were less likely to survive, possibly due to assignment to lower decks or roles with more risk.

Stowaways:

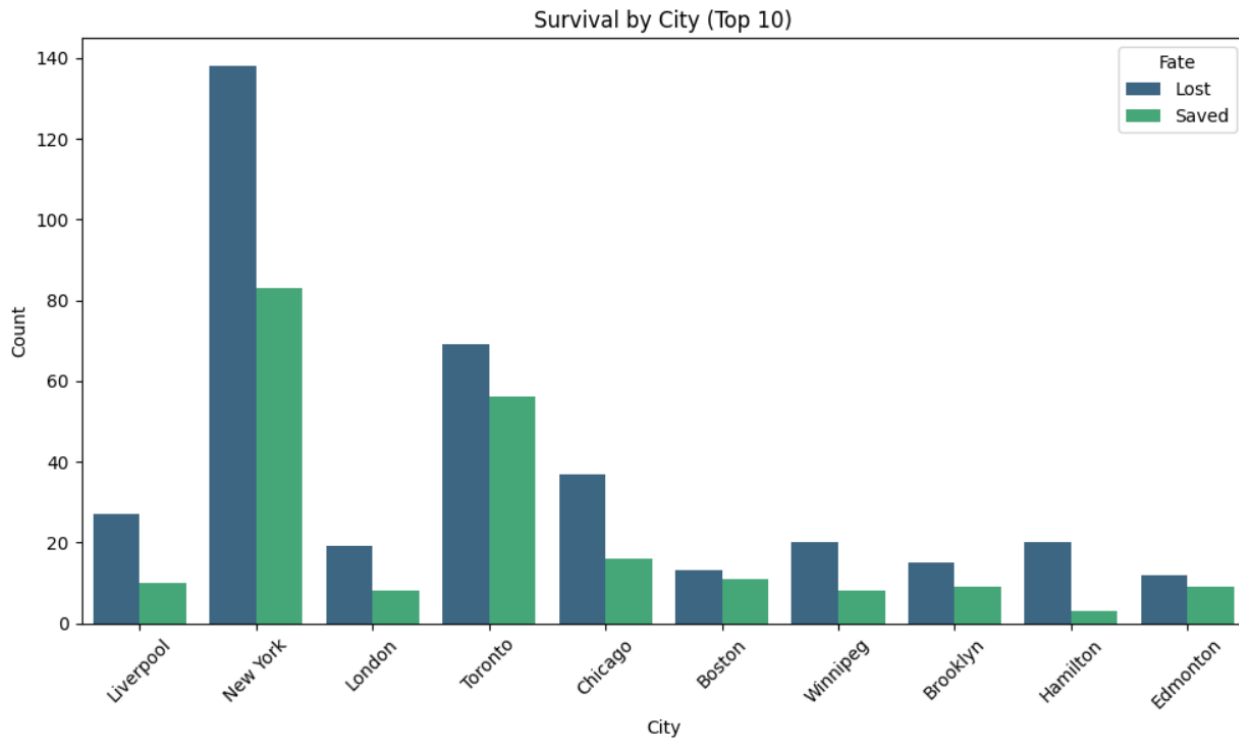
- **Zero** stowaways were saved for both males and females.
- Stowaways likely had **low visibility and priority**, and may have been in inaccessible or hidden locations during the disaster.

7) This chart is useful because it visually compares the number of lost versus saved individuals across the top 10 cities, allowing for a quick understanding of the relative impact of the event in different locations. It highlights cities with the highest numbers and shows the balance between survival outcomes in each.

```
top_cities = data['City'].value_counts().nlargest(10).index
data_top_cities = data[data['City'].isin(top_cities)]
```

```
plt.figure(figsize=(10, 6))
```

```
sns.countplot(x='City', hue='Fate', data=data_top_cities, palette='viridis')
plt.title("Survival by City (Top 10)")
plt.xticks(rotation=45)
plt.xlabel("City")
plt.ylabel("Count")
plt.tight_layout()
plt.show()
```



- The chart is a grouped bar chart visualizing the survival rate by city for the top 10 cities with the most records in the
New York had the highest number of both lost and saved individuals among the top 10 cities. However, the number of lost individuals significantly outweighs the number of saved individuals in New York.
- Toronto also shows a substantial number of both lost and saved individuals. In Toronto, the number of saved individuals is closer to the number of lost individuals compared to New York.
- Liverpool and London had a relatively lower number of both lost and saved individuals compared to New York and Toronto, but in both cities, the number of lost individuals is higher than the number saved.
- Chicago, Boston, Winnipeg, Brooklyn, Hamilton, and Edmonton all show relatively smaller numbers of both lost and saved individuals compared to the top few cities. In most of these cities, the number of lost individuals is still higher than the number saved, although the difference varies.